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Roll No

EE-801-GS
B.E. VIII Semester
Examination, December 2020
Grading System (GS)
Control Systems
Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Derive the transfer function of a armature controlled d.c. motor.
b) Explain the principle of stepper motors.
2. a) Derive the expression for rise time, peak time and maximum over shoot of a second order system.
b) Explain the principle of operation of AC servomotor.
3. a) Plot the magnitude and phase angle plot of the following:
i) K (constant)
ii) s
iii) $\frac{1}{s}$
b) Sketch the Bode plot for the transfer function.

$$G(s) = \frac{50}{s(s+1)(s+2)}$$

Determine:

- i) Gain cross over frequency
- ii) GM and PM
- iii) Stability

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4. A unity feedback control system has $G(s) = 10 / s(s+1)(s+2)$. Draw Nyquist plot and comment on closed loop stability.
5.
 - a) Derive the expression for second order control system when input is subject to unit step.
 - b) The open loop transfer function of unity feedback system is given by $G(S) = K/s(s+1)(s+2)$. Find the minimum value of K for which the steady state error is less than 0.1 for unit ramp input.
6.
 - a) With the help of a neat block diagram explain the working of PI controller.
 - b) Explain with a neat sketch explain the working of “Potentiometer” error detector.
7. Construct Bode plot for the system whose open loop transfer function is given by below and determine
 - a) The gain margin
 - b) The phase margin and
 - c) The closed loop stability.
8.
 - a) The transfer function of a unity feedback control system is given by $\frac{C(s)}{R(s)} = \frac{25}{s^2 + 5s + 25}$. Calculate $w_n, z_u, w_d, \%m_p$.
 - b) Apply Routh criterion to determine the stability of the system having characteristics equation as $s^5 + 1.5s^4 + 2s^3 + 4s^2 + 5s + 10$
